Detailed Game Specification:
Dino Jump

Course: *COMP 2659, Fall 2026*
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1. General Game Overview

This game is based on the game in a Google Chrome tab when you are not connected to the internet. The aim of the game is to get the highest score by "running" as long as possible while avoiding cacti by jumping over them by pressing the spacebar.

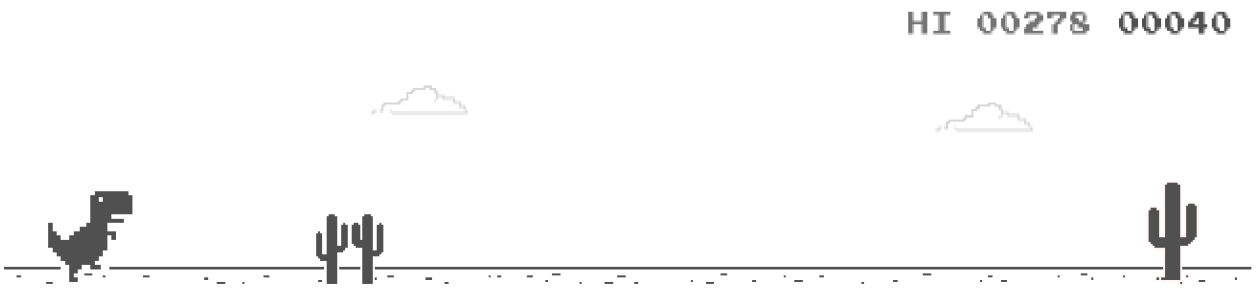


Figure 1. Gameplay screenshot Credit: chrome://dino

2. Game Play Details for Core 1-Player Version

Objectives and Rules

The start screen will be composed of a mainly empty screen with only the dino on the leftmost side and text beside the dino prompting the player to start by pressing the spacebar, as shown in Figure 2.

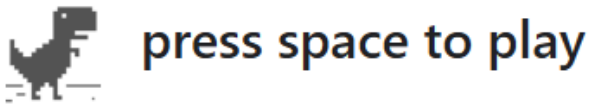


Figure 2. Start menu mockup Credit: chrome://dino

The goal is to survive as long as possible. The player receives one point for every 100ms (every 7 frames) they survive, which is displayed by the score counter in the top right of the screen. The player uses the space bar to jump, and the escape key (esc) to prematurely end the game. If the player for some reason allows the dino to collide with a hazard, the player loses, and the player is sent to the "Game Over" screen.

Objects

Object or Object Type Name	**Properties**	**Behaviour(s)**	**Graphical Image**
Dino	• X coordinate • Y coordinate • X velocity • Y velocity • Width • Height	• Jump	 32x32 px
Cactus	• X coordinate • Y coordinate • X velocity • Y velocity • Width • Height	• Move	 32x32 px
Floor/Ground	• Y coordinate	Acts as a reference point for the dino to "run" on and for the cacti to move across	
Player's score	• Value	• Increment	

Physics

The physics of this game is shown from the perspective of the dino, so the dino will remain at a set position on the left side of the screen.

For the dino's jump physics, the dino will be given an initial vertical velocity for a jump, and the dino's updated height will be determined by the formula "$h = h_0 + vt$" where **h** is the dino's new y-coordinate, $\mathbf{h_0}$ is the dino's initial height relative to the ground, **v** is the dino's current (vertical) velocity, and **t** is the time step in seconds. The dino's next vertical velocity will be determined by the formula "$v = v_0 + at$" where **v** is the dino's updated velocity, $\mathbf{v_0}$ is the dino's initial vertical velocity, **a** is a constant acceleration value, and **t** is the time step in seconds.

The obstacles will move from the right side of the screen to the left side of the screen at a set horizontal velocity of 2 pixels per second. Each obstacle will be given a constant vertical velocity of 0 and a horizontal velocity of 2 pixels per second to the left.

The Dino's "height" will be 2 meters tall. When the Dino jumps, it will be given an initial vertical velocity of 6m/s, the cacti will be 1.5 meters tall, and acceleration due to gravity will be $10m/s^2$.

Asynchronous (Input) Events

Event Name	**Triggering Input Event**	**Description**
SpacePress	Space bar is pressed	If the dino is firmly on the ground, it gets a vertical velocity to jump. The dino will take off from the ground with a set vertical velocity and slowly fall back down towards the ground.
Quit	Esc key is pressed	Gameplay automatically ends regardless of whether or not the player(s) have lost

Synchronous (Timed) Events

Event Name	**Trigger Timing**	**Description**
MoveCactus	Every 1-2 frames	Cactus moves from right to left by 2 pixels
ScorePlayer	Every 100ms (every 7 frames)	Player's score is incremented by 1 point

Condition-Based (Cascaded) Events

Event Name	**Triggering Condition**	**Description**
MoveDino	SpacePress	Dino is moved according to its vertical position and velocity
DinoDies	Dino hits an obstacle (cactus)	The dino runs into an obstacle, causing it to die and triggering a game over.
GameOver	When the "DinoDies" event is triggered	Game is terminated and displays the game over screen
GameQuit	Asynchronous "Quit" event is triggered	Game is terminated forcefully for all players



Figure 3. Game over Credit: chrome://dino

Hypothetical Gaming Session

The player will press the spacebar on the game start screen, and then the game will start. As the game goes on, the cacti will move towards the dino, and the player will press the spacebar, which will make the dino jump over and avoid the cacti; while this is happening, the score will increase incrementally. The player will continue this gameplay loop until they mess up and the dino touches the cacti, resulting in a game over. The player will start over and retry in an attempt to beat their previous score.

3. Game Play Details for Core 2-Player Version

Player 1 and Player 2 will be shown on the screen together, with Player 1 shown on the top 200 pixels and Player 2 shown on the bottom 200 pixels. The game will continue infinitely with each player jumping over their own cactus while trying to outlast the other player. Whichever player manages to go longer without dying will be the winner. The cacti will appear at the same x-axis position for each player in order for it to be fair for each player.

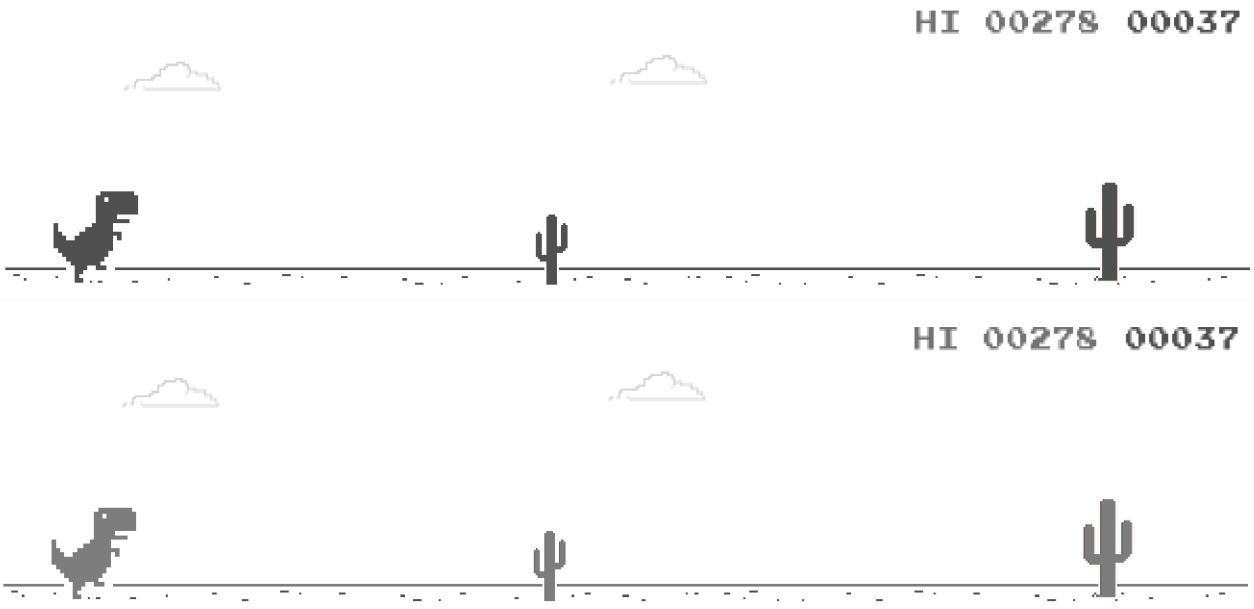


Figure 4. Mockup of a 2-player version Credit: chrome://dino

4. Sound Effects

Sound Effect Name	Brief Description	Event which Triggers Playback
DinoDeathSound	A staticky 2-tone sound effect where the first tone is higher than the second tone. Static-sounding 'ba bum'.	DinoDies
JumpSound	A 1245hz tone (E ♭ 6) that lasts 300ms (21 frames)	SpacePress

The background music will be a 4-note chill electronic retro synthwave beat on loop.

5. Additional Features (Time Permitting)

1) Add pterodactyls as an additional hazard for the player(s) to avoid that fly at a set height. In addition, add a ducking "animation" for the dino to avoid the pterodactyls

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2) Challenge mode, where the obstacles move towards the dino much faster, along with there being much tighter jumps requiring extra precision, such as having multiple cacti back to back or requiring the player to try and jump much earlier in order to be able to immediately make another jump in order to jump over the next cactus.

3) The time of day will change every 1000 points gained. When the player hits odd 1000s of points (e.g. 1000, 3000, 5000, etc.), the sky will turn black, while the ground, dino, and cactus turn white. When the player hits even 1000s of points (e.g. 2000, 4000, 6000, etc.), the screen will turn back to normal (i.e., the colour scheme for the starting screen)

4) Multiple and varying cactus shapes and styles.